Medida de Eficiência - Cálculo II

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1ª)

ME de Cálculo II

1ª) a) $f(x;y) = 4 - x^2 - 2y^2$

$f_x = -2x$ | $f_y = -4y$

$f_x(1;1) = -2(1) = -2$ // | $f_y(1;1) = -4(1) = -4$ //

b) $W(x;y) = xy^2e^{xy}$ $\quad u' = 2xy \quad v' = xe^{xy}$

$W_y = 2xye^{xy} + xy^2 \cdot xe^{xy}$

$= e^{xy}(2xy + x^2y^2)$

$W_y = xye^{xy}(2 + xy)$ //

c) $W = x^2y^3 \cdot sen(z) + e^{xz}$

$W_x = 2xy^3 \cdot sen(z) + ze^{xz}$

$W_y = 3x^2y^2 \cdot sen(z) + 0$

$W_z = x^2y^3 \cdot cos(z) + xe^{xz}$

2ª e 3ª)

2º) $f(x;y) = x^3 + x^2y^3 - 2y^2$

$f_x = 3x^2 + 2xy^3$

$f_{xx} = 6x + 2y^3$

$f_{xy} = 6xy^2$

$f_y = 3x^2y^2 - 4y$

$f_{yy} = 6x^2y - 4$

$f_{yx} = 12xy$

3º) a) $z = 5xy$

$z_x = 5y$ | $z_y = 5x$

$z_{xx} = 0$ | $z_{yy} = 0$

$z_{xx} + z_{yy} = 0$

$0 + 0 = 0$

Satisfaz

b) $z = e^x \operatorname{sen}(y)$

$z_x = e^x \operatorname{sen}(y)$ | $z_y = e^x \cdot \cos(y)$

$z_{xx} = e^x \operatorname{sen}(y)$ | $z_{yy} = -e^x \cdot \operatorname{sen}(y)$

$e^x \operatorname{sen}(y) + (-e^x \operatorname{sen}(y)) = 0$

Satisfaz

4ª)

4:) $\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt} + \frac{dw}{dy} \cdot \frac{dy}{dt}$ $\qquad W = x^2 + y^2$

$\frac{dw}{dx} = 2x$ | $\frac{dw}{dy} = 2y$

$\frac{dx}{dt} = -\operatorname{sen}(t)$ | $\frac{dy}{dt} = \cos(t)$

$\frac{dw}{dt} = (2x)(-\operatorname{sen}(t)) + (2y)(\cos(t))$

$\frac{dw}{dt} = 2x \cdot (-0) + 2y \cdot (-1)$

$\frac{dw}{dt} = 0 - 2y = -2y$ //

5ª)

5ª) $u = x^4 y + y^2 z^3$ $\qquad \frac{du}{ds} = \frac{du}{dx}\cdot\frac{dx}{ds} + \frac{du}{dy}\cdot\frac{dy}{ds} + \frac{du}{dz}\cdot\frac{dz}{ds}$

$\frac{du}{dx} = 4x^3 y$ | $\frac{du}{dy} = x^4 + 2yz^3$ | $\frac{du}{dz} = 3y^2 z^2$

$x = rse^t$
$y = rs^2 e^{-t}$
$z = sr^2 \operatorname{sen}(t)$

$\frac{dx}{ds} = re^t$ | $\frac{dy}{ds} = 2re^{-t}s$ | $\frac{dz}{ds} = r^2 \cdot \operatorname{sen}(t)$

$r = 2,\ s = 1,\ t = 0$

$x = 2\cdot 1\cdot e^0 = 2$; $y = 2(1)^2 e^{-0} = 2$; $z = 1(2)^2 \operatorname{sen}(0) = 0$

$\frac{dx}{ds} = 2\cdot e^0 = 2$ | $\frac{dy}{ds} = 2(2)e^0(1) = 4$ | $\frac{dz}{ds} = (2)^2 \cdot \operatorname{sen}(0) = 0$

$\frac{du}{dx} = 4(2)^3 2 = 64$ | $\frac{du}{dy} = (2)^4 + 2(2)(0)^3 = 16$ | $\frac{du}{dz} = 3(2)^2(0)^2 = 0$

$\frac{du}{ds} = 64\cdot 2 + 16\cdot 4 + 0\cdot 0$

$$\boxed{\frac{du}{ds} = 128 + 64 = 192}$$